Functions, notation and logarithms: why AI adds log-probabilities

A language model gives every possible next word a probability. Why does training work with the logarithm of that probability?

Mathematics for AI and Machine Learning | Session 1 of 18 | Foundations
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Read sigma notation and function notation, and evaluate both by hand

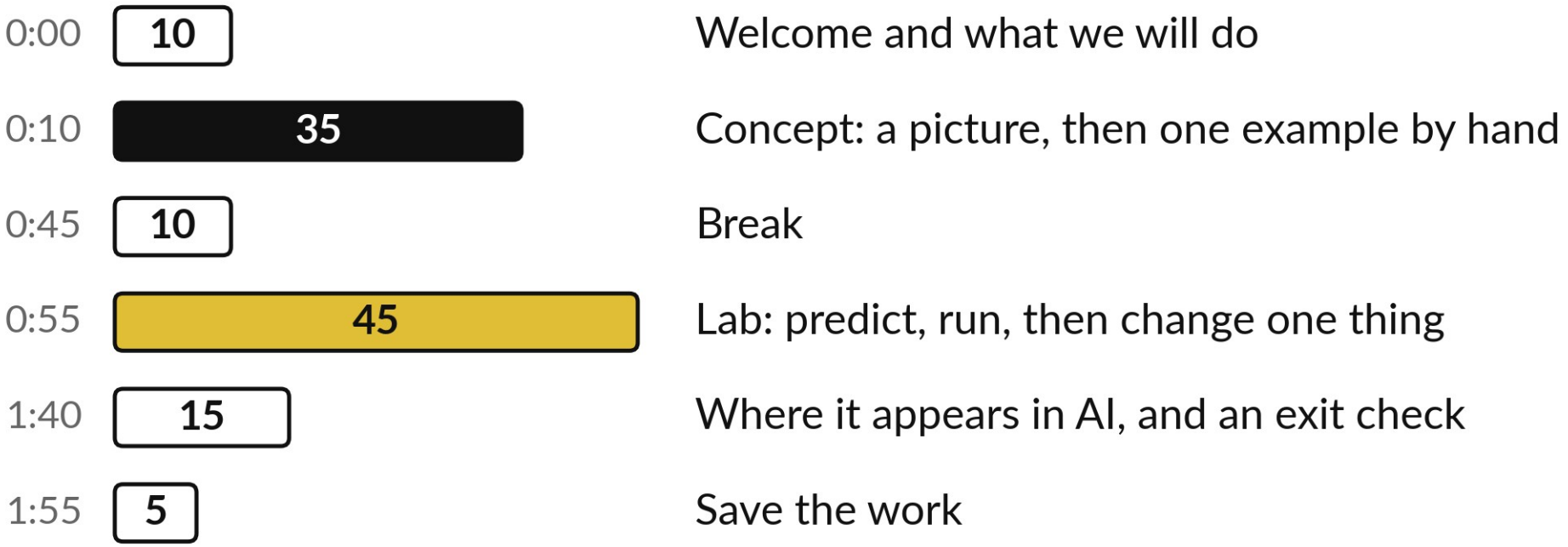


Use exponentials and logarithms, and the three log rules, to turn a product into a sum



Compute the log-probability of a word and of a sentence in Python and in a spreadsheet

Your two hours: sums, functions and logarithms, then the lab, then AI



Ideas first. Then the lab, in Python or in a spreadsheet. Then where it shows up in AI.

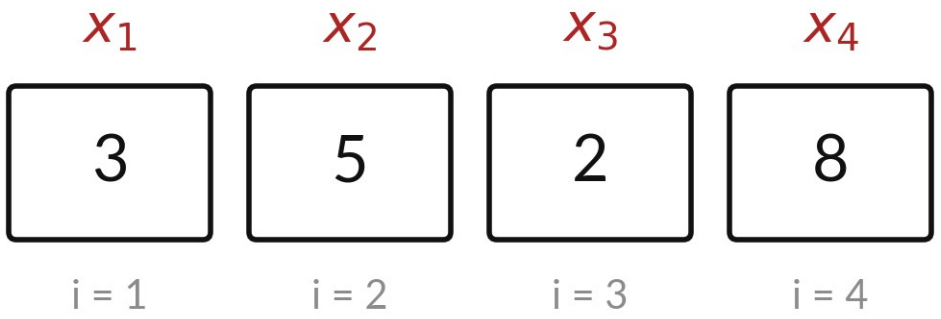
Words we use today

	Word	What it means
	**Sigma notation**	The symbol Σ means add up. Sigma notation is a short way to write a long sum.
	**Function**	A rule that turns each input number into one output number, written f(x).
	**Exponential function**	A function where the input is a power, such as 2^x (2 to the power x). It grows by repeated multiplying.
	**Logarithm (ln)**	The inverse of an exponential. The natural logarithm ln uses the base e ≈ 2.71828.
	**Log-probability**	The natural logarithm of a probability. It is zero or negative, because a probability is at most 1.

The sigma symbol is a short way to write a long sum

$$\sum_{i=1}^{4} x_i$$

Add up the x-values from i = 1 to i = 4



3 + 5 + 2 + 8 = 18

Sigma notation: the capital Greek letter Σ means add up.

Below Σ, i = 1 is where to start. Above Σ, the 4 is where to stop.

Each box is one value, x with its index i.

The values 3, 5, 2 and 8 add up to 18.

Sigma notation in action: the five next-word probabilities add up to 1

General sum

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n$$

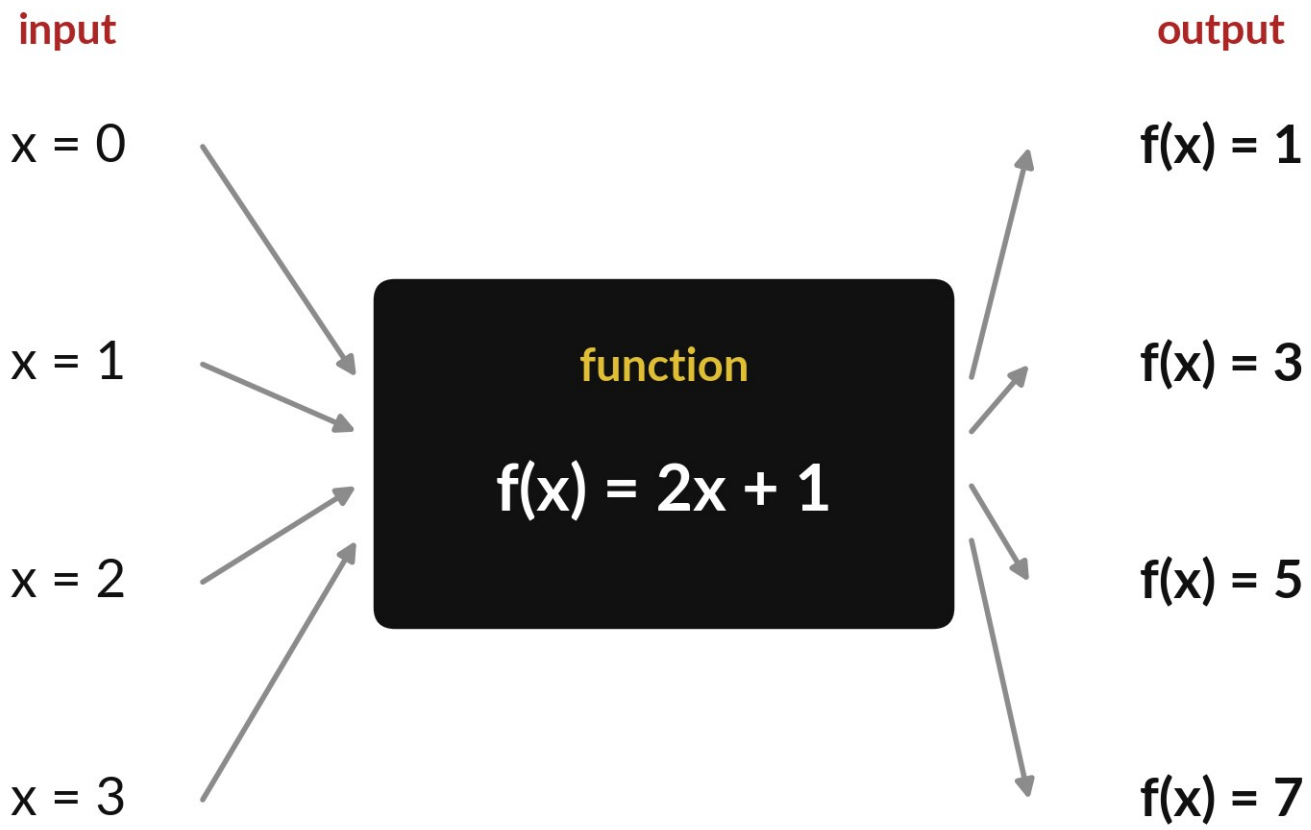
Our five words

$$\sum_{i=1}^{5} p_i = 0.62 + 0.18 + 0.10 + 0.06 + 0.04 = 1.00$$

Symbol	What it means	In our data
Σ (sigma)	Add up everything after it.	a sum of five numbers
i	The index: the position in the list (1, 2, 3 ...).	i = 1 to 5
n	How many values there are.	n = 5 words
p_i	The probability of word i after "The capital of Uganda is".	p_1 = 0.62 for Kampala

Illustrative (made-up numbers): Kampala 0.62, Entebbe 0.18, a 0.10, the 0.06, Nairobi 0.04.

A function turns each input number into exactly one output number



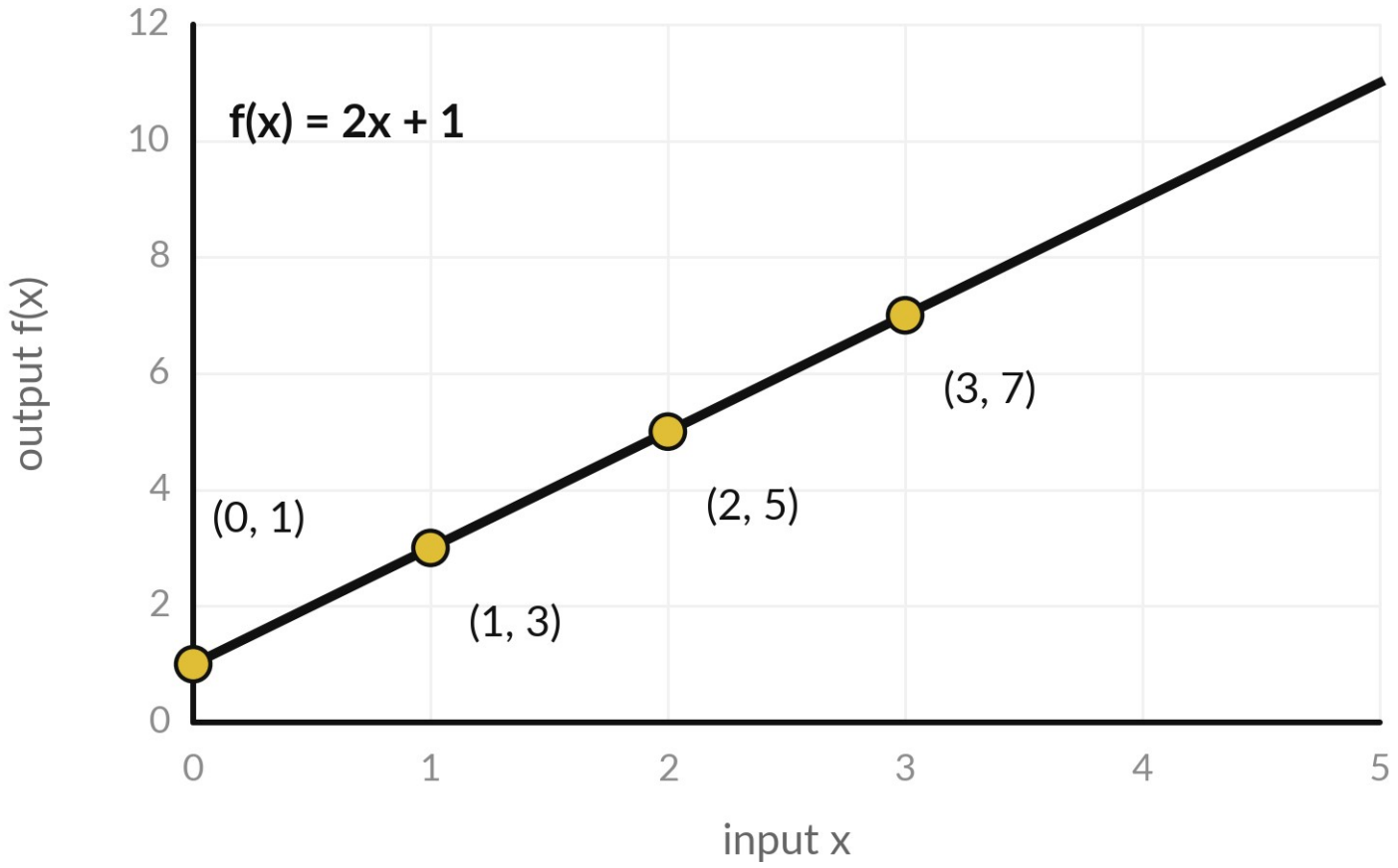
A function is a rule. Give it an input x and it returns one output, written f(x).

Here the rule is f(x) = 2x + 1: double the input, then add 1.

Put in x = 3. Double it to get 6, add 1, and f(3) = 7.

The same input always gives the same output.

A graph draws every input and its output as a point



A graph plots each pair (x, f(x)) as a point. Joining the points draws the curve.

Across: the inputs x. Up: the outputs f(x).

Each gold dot is one pair, such as x = 3 and f(3) = 7.

Each step of 1 in x adds 2: a straight line.

The function f(x) = wx + b is the building block of machine learning

A linear function

$$f(x) = w\,x + b$$

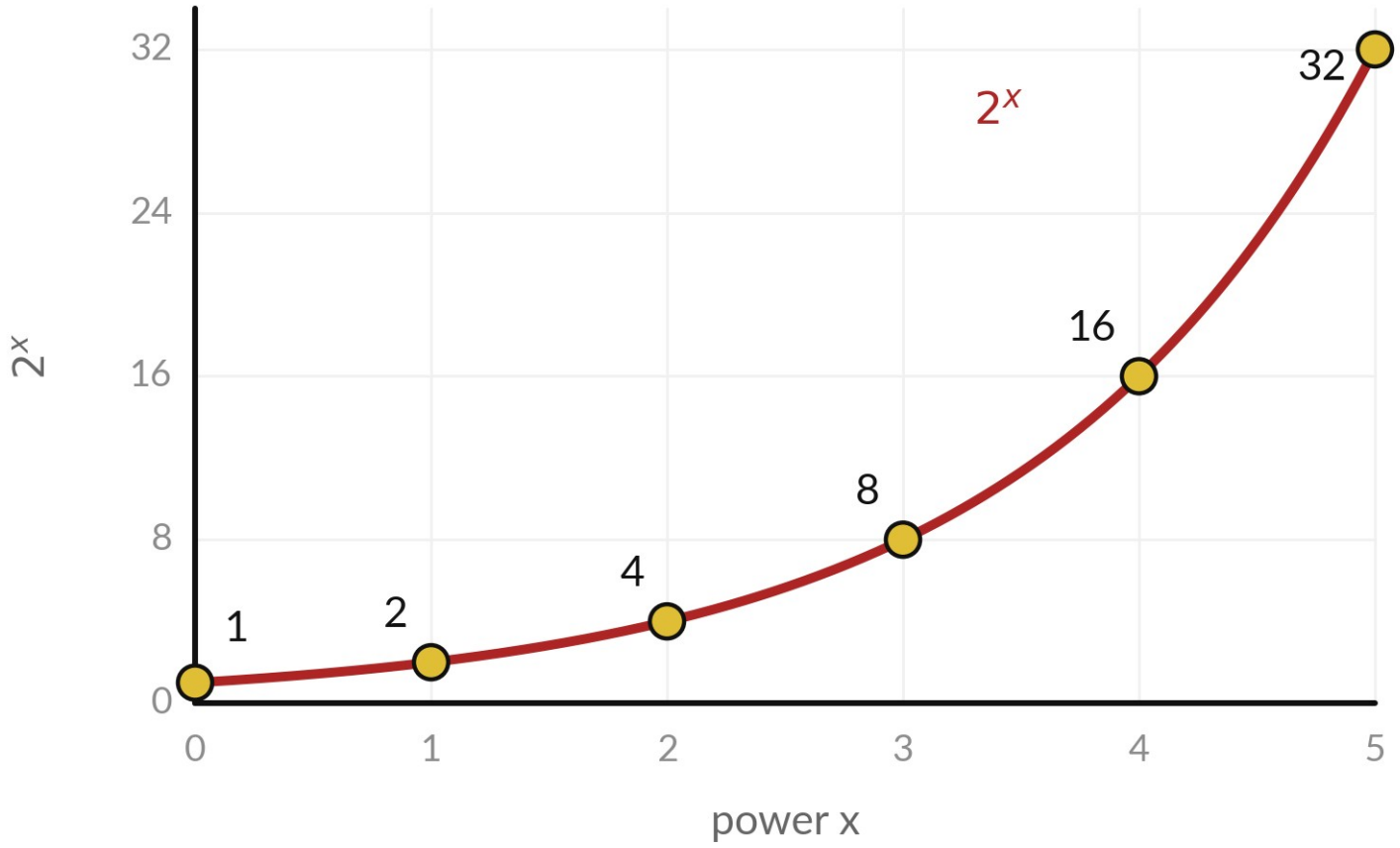
With w = 2 and b = 1, at x = 3

$$f(3) = 2 \times 3 + 1 = 7$$

Symbol	What it means	In our data
x	The input.	x = 3
f(x)	The output for the input x.	f(3) = 7
w	The weight: how much the output grows when x grows by 1. It is also called the slope.	w = 2
b	The bias (intercept): the output when the input is 0.	b = 1

w and b are parameters: numbers a model learns. In this course b always means the bias.

An exponential function multiplies again and again, so it grows very fast

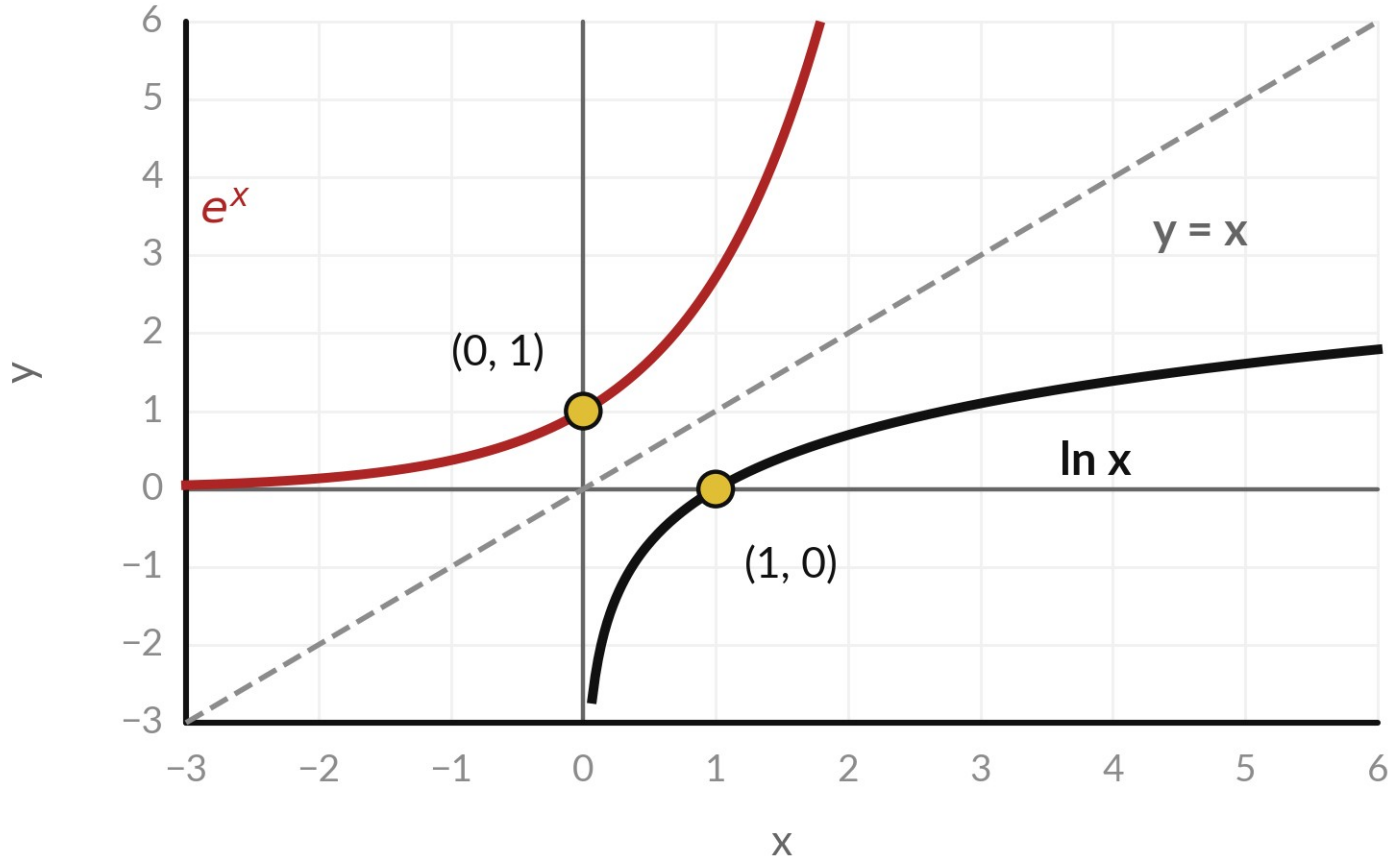


Exponential function: the input is the power. 2^x means x copies of 2 multiplied together: $2^3 = 2 \times 2 \times 2 = 8$.

$2^1 = 2$, $2^2 = 4$, $2^3 = 8$: each step right doubles. Each step left halves, so $2^0 = 1$.

The base e ≈ 2.71828 gives e^x: EXP(x), np.exp(x). e^x is always above 0: $e^{-1} \approx 0.37$.

A logarithm undoes an exponential: it answers "which power gives this number?"



The logarithm is the inverse of the exponential. If $2^3 = 8$, then log base 2 of 8 is 3.

The curves e^x and ln x are mirror images across the dashed line y = x.

ln x is the natural logarithm: the logarithm with base e.

ln 1 = 0, because $e^0 = 1$. And ln x exists only for x above 0.

The logarithm as a formula: the answer x is the power that gives y

Definition, base a

$$\log_a(y) = x \iff a^x = y$$

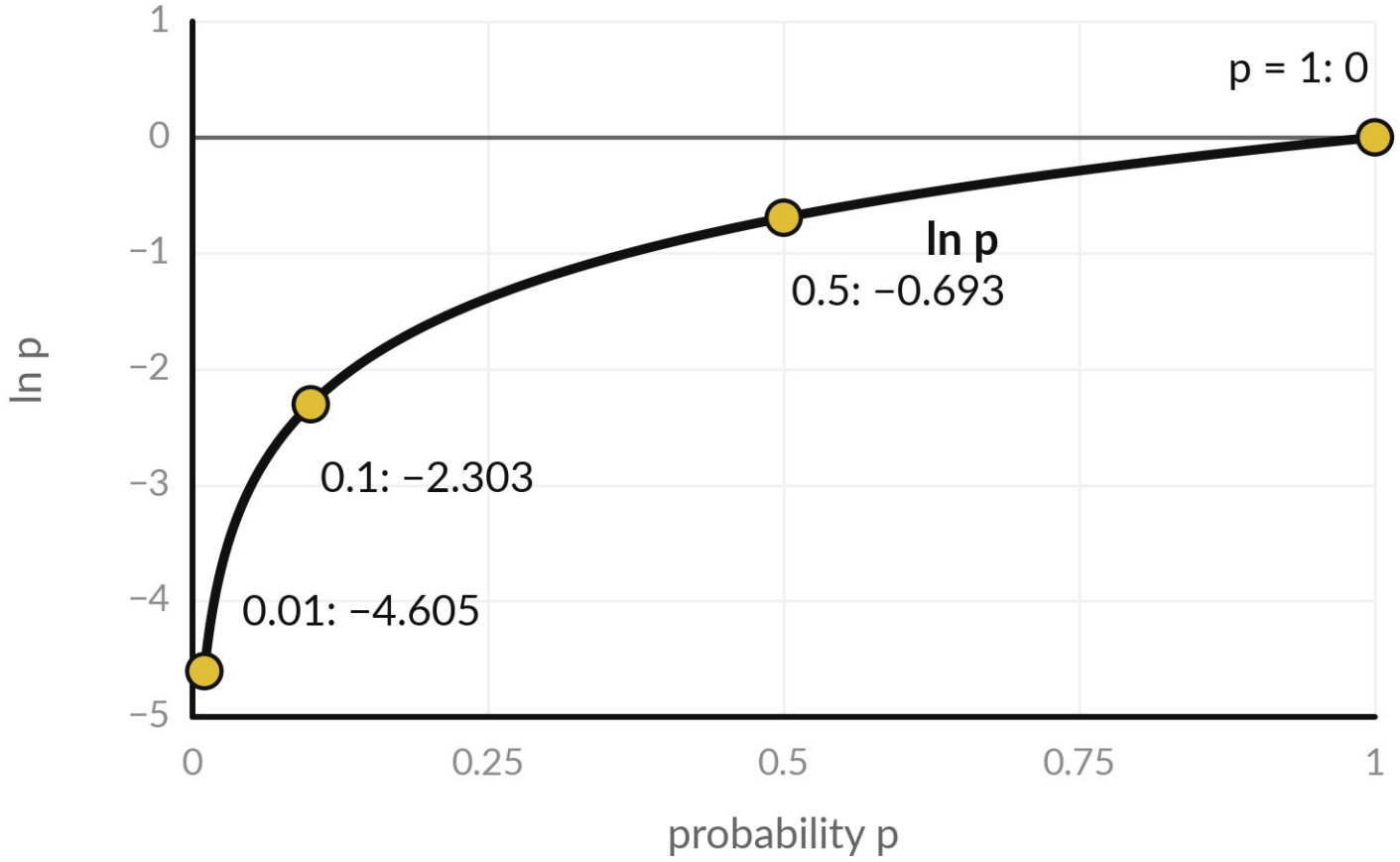
Examples

$$\log_2 8 = 3, \quad \log_{10} 1000 = 3, \quad \ln e = 1$$

Symbol	What it means	In our data
a	The base: the number that is multiplied by itself. Common bases are 2, 10 and e ≈ 2.71828. (b stays the bias.)	a = 2 in $\log_2 8 = 3$
ln	The natural logarithm: the logarithm with base e.	ln 6 = 1.7918
y	The number we take the logarithm of. It must be above 0.	y = 8 in $\log_2 8 = 3$
x	The answer: the power that a must be raised to, to give y.	x = 3 in $\log_2 8 = 3$

Python: np.log is natural, np.log10 is base 10. Spreadsheet: LN is natural, LOG10 is base 10.

A probability lies between 0 and 1, so its natural logarithm is zero or negative



A certain event (p = 1) has ln p = 0. The smaller the probability, the more negative ln p.

p = 1 gives 0. p = 0.5 gives −0.693.

Minus ln p grows as p shrinks.

For the right word it is the loss (the number training makes small).

Log rules: a logarithm turns multiplying into adding

Product rule $$\ln(u \times v) = \ln u + \ln v$$

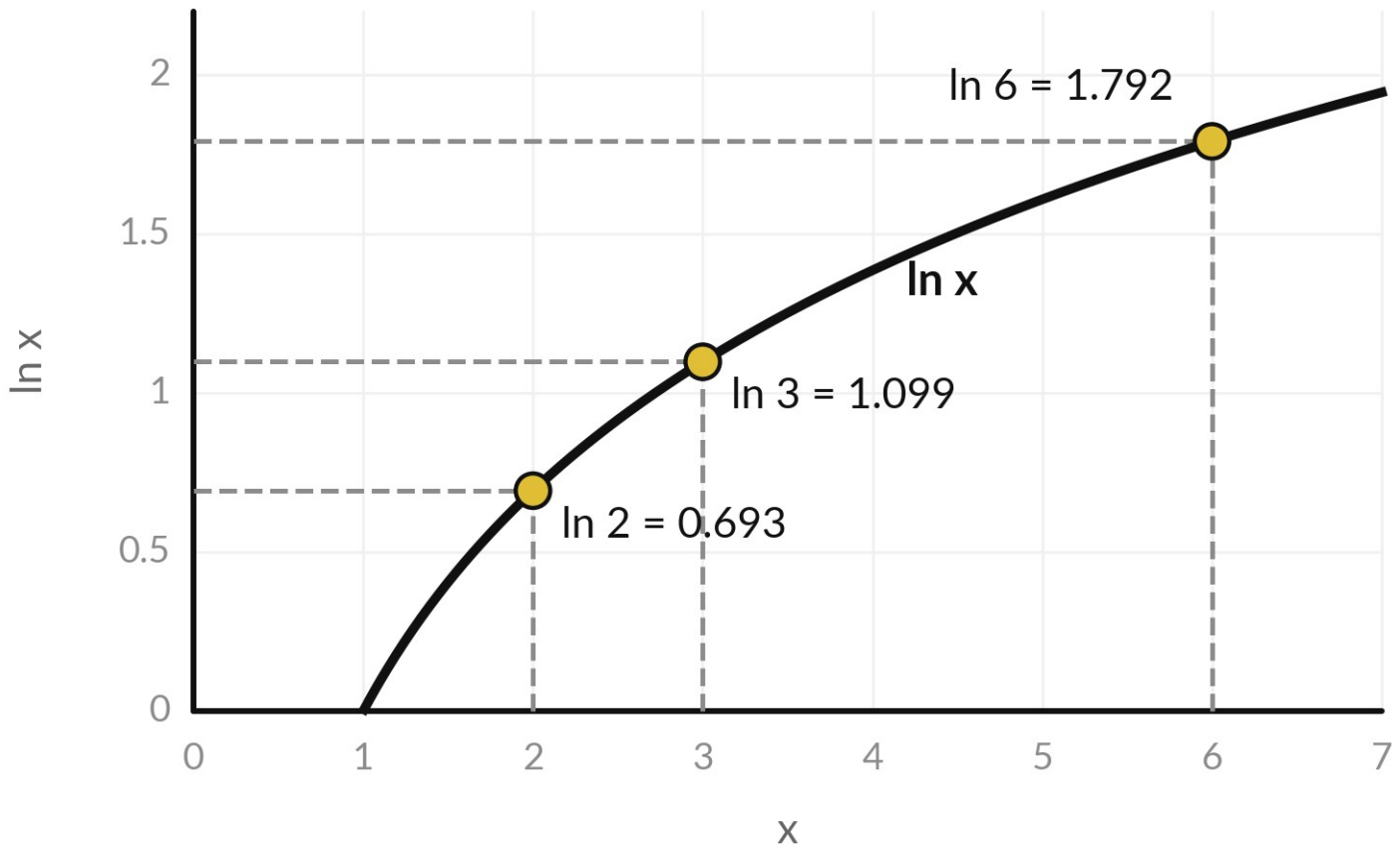
Quotient rule $$\ln\left(\frac{u}{v}\right) = \ln u - \ln v$$

Power rule $$\ln(u^k) = k \ln u$$

Symbol	What it means	In our data
u and v	Any two numbers above 0.	u = 2, v = 3
k	Any number, used as a power.	k = 2
Product rule	ln 6 = ln 2 + ln 3	1.7918 = 0.6931 + 1.0986
Power rule	ln 9 = ln(3^2) = 2 ln 3	2.1972 = 2 × 1.0986

Check the quotient rule: ln(6 ÷ 2) = ln 3 = 1.0986, and ln 6 − ln 2 = 1.7918 − 0.6931 = 1.0986.

Check the product rule on the curve: ln 2 plus ln 3 equals ln 6



Product rule check with u = 2 and v = 3: ln 2 + ln 3 should equal ln 6.

ln 2 = 0.6931 and ln 3 = 1.0986. Their sum is 1.7918.

The height of the curve at x = 6 is also 1.7918, because 2 × 3 = 6.

Heights on the curve add when the inputs multiply.

A sentence has the probability of all its words multiplied together

The lab sentence, one word at a time (illustrative: made-up probabilities)

The	×	capital	×	of	×	Uganda	×	is	×	Kampala
p = 0.3		p = 0.2		p = 0.5		p = 0.4		p = 0.7		p = 0.62
ln p = −1.204		ln p = −1.609		ln p = −0.693		ln p = −0.916		ln p = −0.357		ln p = −0.478

Multiply the probabilities: 0.3 × 0.2 × 0.5 × 0.4 × 0.7 × 0.62 = 0.005208

Add the 6 ln p values: −5.2576

Multiply the six word probabilities for the sentence. Add their logarithms for its log-probability.

The log of a product is the sum of the logs: ln P is the sum of the ln p_i

$$P = \prod_{i=1}^{n} p_i \quad \Longrightarrow \quad \ln P = \sum_{i=1}^{n} \ln p_i$$

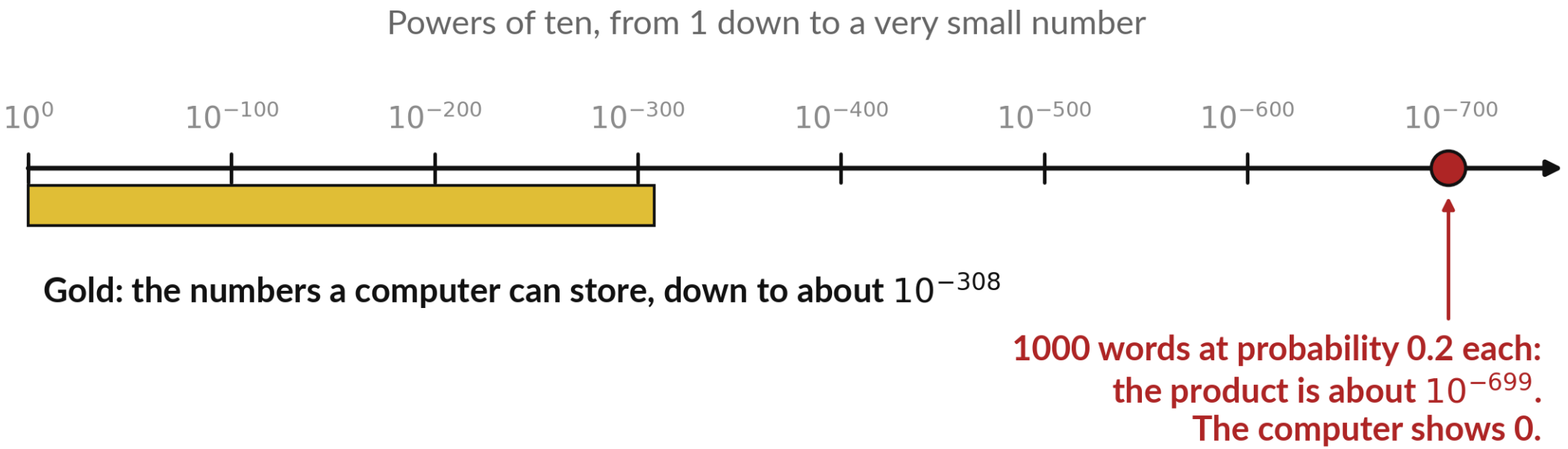
Our six words

$$\ln P = -1.204 - 1.609 - 0.693 - 0.916 - 0.357 - 0.478 = -5.2576$$

Symbol	What it means	In our data
Π (capital pi)	Multiply everything that comes after it. It is the product symbol.	six probabilities
p_i	The probability the model gave to word i, knowing the words before it.	0.3, 0.2, 0.5, 0.4, 0.7, 0.62
P	The probability of the whole sentence.	0.005208
ln P	The log-probability of the sentence: zero or negative.	−5.2576

Plain average per word: 5.2576 ÷ 6 = 0.8763 nats, the training loss (cross-entropy, Session 17).

A long product of probabilities shrinks to zero, but the sum of the logarithms stays usable



The sum of the logarithms is 1000 × ln 0.2 = −1609.4. That is an ordinary number.

A computer stores numbers down to about 10^{-308}. Below that, a product rounds to 0 (underflow).

Common mistake: ln(u + v) differs from ln u + ln v

Wrong

$\ln(2 + 3) = \ln 2 + \ln 3$

That gives 1.7918, but ln 5 is 1.6094.

np.log is base 10.

Right

$\ln(2 + 3) = \ln 5 = 1.6094$.

$\ln(2 \times 3) = \ln 2 + \ln 3 = 1.7918$.

np.log is the natural logarithm (base e).

The log rules are for products only.

Read the recipe before you run it: sums, logarithms and exponentials

```
p = np.array([0.62, 0.18, 0.10, 0.06, 0.04])
total = np.sum(p)
neg_log_p = -np.log(p)
entropy = np.sum(p * neg_log_p)
sentence_p = np.prod(word_p)
back = np.exp(np.log(6))
```

In plain words

Line 1: the five probabilities in a list called an array.

Line 2: add them up, which is the sigma sum.

Line 3: minus ln of each one, all at once.

Line 4: each p times its minus ln p, added up: the entropy, 1.133 nats (Session 17). Spreadsheet: SUMPRODUCT.

Line 5: multiply all six word probabilities. Spreadsheet: =PRODUCT(B27:B32).

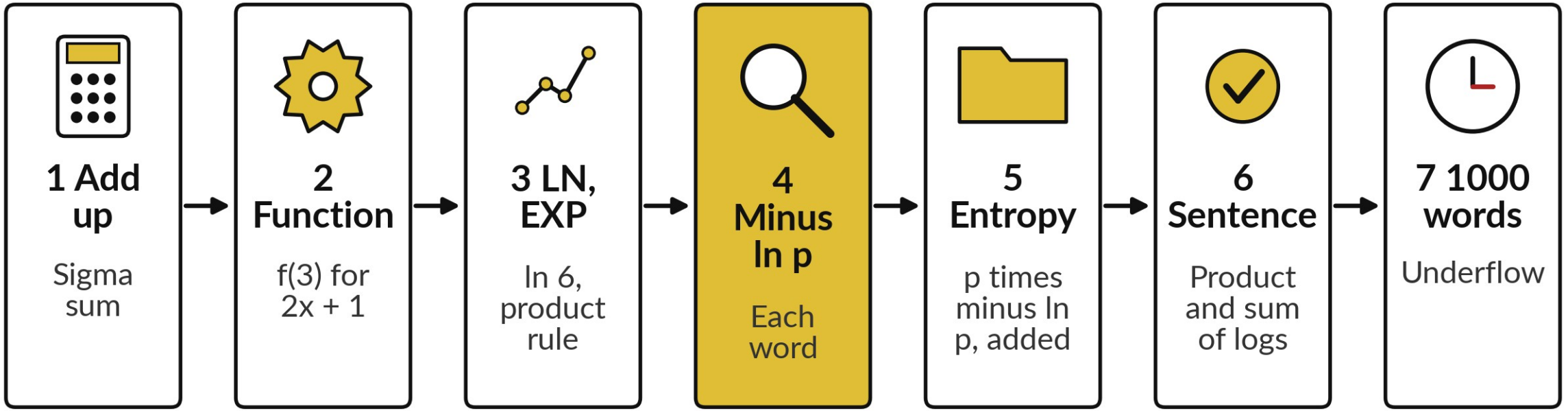
Line 6: EXP undoes LN, so you get 6 back.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has seven steps, and each one uses a number from the slides



Predict first, run, read, then change one thing. Steps 6 and 7 are the sentence of six words, then the 1000-word test.

Check your numbers before you move on

	What you compute	You should get
	**Sum of the five probabilities**	1.00
	**f(3) for f(x) = 2x + 1**	7
	**ln 6, and ln 2 + ln 3**	1.7918
	**Minus ln p for Kampala (p = 0.62)**	0.478
	**Entropy: sum of p times minus ln p, five words**	1.133 nats
	**Six word p: product, then sum of ln**	0.005208 and −5.2576
	**Plain average of minus ln p per word, sentence**	0.8763
	**1000 words, p = 0.2: sum of logs**	−1609.44

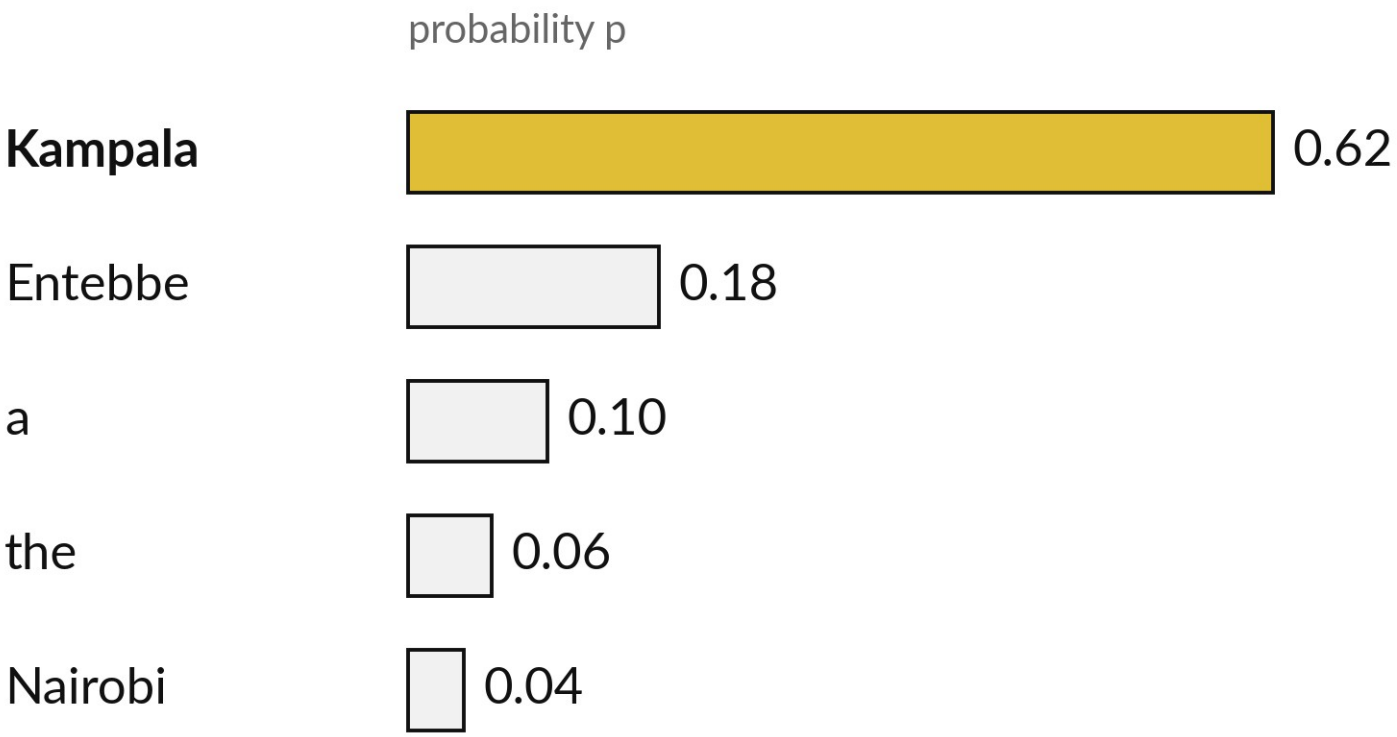
If your number differs, find the step that went wrong. Do not change a number to match.

Four messages you may meet, what they mean and what to do

What you see	What it means	What to do
nan or a RuntimeWarning from np.log, or #NUM! from LN	You took the log of 0 or of a negative number. A logarithm needs a number above 0.	Check the probabilities. None may be 0 or negative.
My log-probability is negative	That is correct. A probability below 1 has a negative natural logarithm.	Use minus ln p when you want a positive number.
The 1000-word product shows 0.0	Underflow: the true value is about 10^{-699}, too small to store.	Use the sum of the logs. That is the point of the step.
#NAME? in a spreadsheet	The function name is misspelt, or your region uses a semicolon between the arguments.	Retype the function. Try ; instead of ,

Where this appears in AI: a probability for every possible next word

The capital of Uganda is …



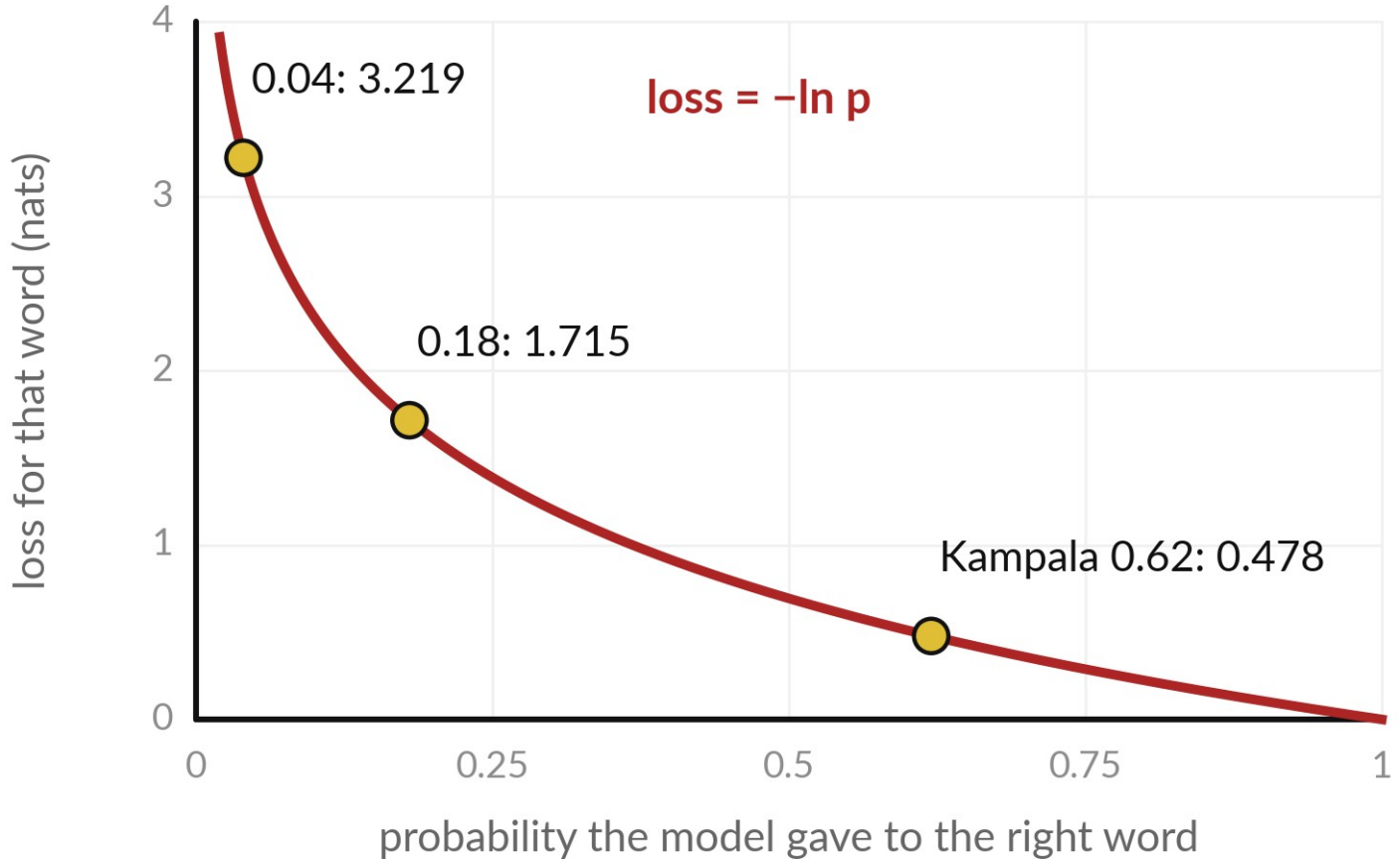
Language models of this kind give every possible next token a probability.

A token is a piece of text, often a word or part of a word.

These numbers are illustrative (made up). They add up to 1.

Vocabulary of open models: Llama 2, 32,000 tokens; Gemma, 256,000.

Where this appears in AI: training uses ln of the right word probability



Training scores the model by minus ln p, where p is the probability it gave the word that really came next.

The right word is Kampala. With p = 0.62 the loss is 0.478.

With only p = 0.04 the loss is 3.219. A low probability on the right word gives a high loss.

Averaged over a text: cross-entropy (Session 17).

Where this appears in AI: check an assistant with your numbers

8 minutes Hands-on activity

Ask any AI assistant: "What is ln 6, and what is ln 2 plus ln 3?"

Use only public maths. Never paste confidential data.

Compare its ln 6 with your lab number, 1.7918.

Ask why the two answers agree. Does the reasoning match the product rule?

Ask the same question again. Does the answer change?

You keep

A note: the assistant answer, your lab number, and whether they agree.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**What is $\Sigma\, p_i$ for i = 1 to 3, when p_1 = 0.5, p_2 = 0.3 and p_3 = 0.2?**	1.0. Sigma means add: 0.5 + 0.3 + 0.2.
	**Fill the gap: ln 12 = ln 3 + ln ?**	ln 4, because 3 × 4 = 12 and the logarithm of a product is the sum of the logarithms.
	**Why do AI programs add log-probabilities instead of multiplying probabilities?**	A product of many small probabilities underflows to 0. A sum of logs stays an ordinary number.

Save your lab file and start your formula sheet

What you keep from today

Keep your saved notebook or workbook, with every check on OK. Keep the one idea of the day: the logarithm of a product is the sum of the logarithms, and that is why AI adds log-probabilities.

Our question today

A language model gives every possible next word a probability. Why does training work with the logarithm of that probability?

Next session

Session 2: Vectors. A row of data becomes a list of numbers that we can add, scale and compare.

Take away

Your lab file
Saved, all checks OK

The idea
$\ln(u \times v) = \ln u + \ln v$

Formula sheet
Σ, $f(x) = wx + b$, log rules

Next session
Vectors